

Absurdly Accurate Clock

Pico 2 / RP2350 pin-assignment reference

Assembly: Populate both 20-pin header rows. Top/component view, USB at top: pin 1 upper left; pin 20 lower left; pin 21 lower right; pin 40 upper right. Physical numbers below are board header pins, not chip pins.

Physical	GPIO	Direction*	Peripheral / function	Target / signal	Wiring / implementation notes
1	GP0	OUT	UART0 TX (F2)	GPS RX	Pico sends GPS configuration / UBX commands.
2	GP1	IN	UART0 RX (F2)	GPS TX	NMEA / UBX serial data from GPS.
4	GP2	IN	GPIO / SIO (F5)	GPS PPS	Timing edge input; configure edge and polarity.
6	GP4	OUT	UART1 TX (F2)	MAX3232 T1IN	T1OUT (RS-232) to Posiflex PD-2200 RX.
7	GP5	IN / reserve	UART1 RX (F2)	MAX3232 R1OUT	Optional return path: display TX to R1IN; else leave spare.
9	GP6	IN	GPIO / SIO (F5)	UI button	Suggested: switch to GND, pull-up, debounce.
11	GP8	IN	SPI1 RX (F1)	ESP32 MISO	Future SPI data from ESP32 to Pico.
12	GP9	OUT	SPI1 CSn (F1) / SIO	ESP32 CS	Active low; software CS (F5) is also possible.
14	GP10	OUT	SPI1 SCK (F1)	ESP32 SCLK	Future SPI clock from Pico.
15	GP11	OUT	SPI1 TX (F1)	ESP32 MOSI	Future SPI data from Pico to ESP32.
16	GP12	OUT	GPIO / SIO (F5)	ESP32 TIME_SYNC	Project-defined timing signal; protocol / pulse width TBD.
17	GP13	TBD / reserve	GPIO / SIO (F5)	ESP32 control / IRQ	Future handshake; direction and polarity not yet assigned.

*Directions are relative to the Pico. SPI assumes Pico controller (master), ESP32 peripheral (slave). ESP32 pin numbers and link protocol remain TBD; NTP service runs on the future ESP32.

Keep free for expansion: GP14 / pin 19; GP15 / 20; GP16 / 21; GP17 / 22; GP18 / 24; GP19 / 25; GP20 / 26; GP21 / 27; GP22 / 29. Also unassigned: GP3 / 5, GP7 / 10, GP26 / 31, GP27 / 32, GP28 / 34.

Ground and interface wiring: Nearby GND pins: 3, 8, 13, 18; other digital GND pins: 23, 28, 38 (pin 33 is AGND). Use common ground and 3.3 V logic at the Pico. Keep RS-232 on the display side of the MAX3232; verify module logic levels and PD-2200 connector wiring before assembly. Power distribution is not specified by this sheet.

Pinmux verified: UART0, UART1 and SPI1 assignments coexist. GP12 / GP13 stay GPIO, despite their other alternate functions. In C++ / PlatformIO, assign GPIO numbers explicitly; do not rely on framework defaults. F1 = SPI, F2 = UART, F5 = SIO. Leave unused reserved signals undriven until their interface is defined.